Internship Report - Day 5

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1 Introduction to Mathematical Modelling

Mathematics the language of universe! Not just poetry, but the eternal truth of every physical system that moves around the time. Mathematical modelling played a major role in shaping the scientific and technological advancements in human history.

Before delving into the description, let us first understand what a model is.

model (n): *a miniature representation of something; a pattern of something to be made; an example for imitation or emulation; a description or analogy used to help visualize something (e.g., an atom) that cannot be directly observed; a system of postulates, data and inferences presented as a mathematical description of an entity or state of affairs.* [5]

Then a **mathematical model (n):** *are representations in mathematical terms of the behaviour of real devices and objects.*

We want to learn how to make or generate such models, how to validate them, how to use them, and how and when their use is limited. But before moving deep — it is worth talking about why we do mathematical modelling.

> *Mathematical modelling is the process of describing a* ***real-world*** *problem in mathematical terms, usually in the form of equations, and then using these equations both to help understand the original problem, and also to discover new features about the problem. Modelling both lies at the heart of much of our understanding of the world, and it allows engineers to design the technology of the future. With modelling we can travel to the edge of the universe, peer into the heart of the atom, and understand the future of our climate.* [4]

1.1 Historical Background

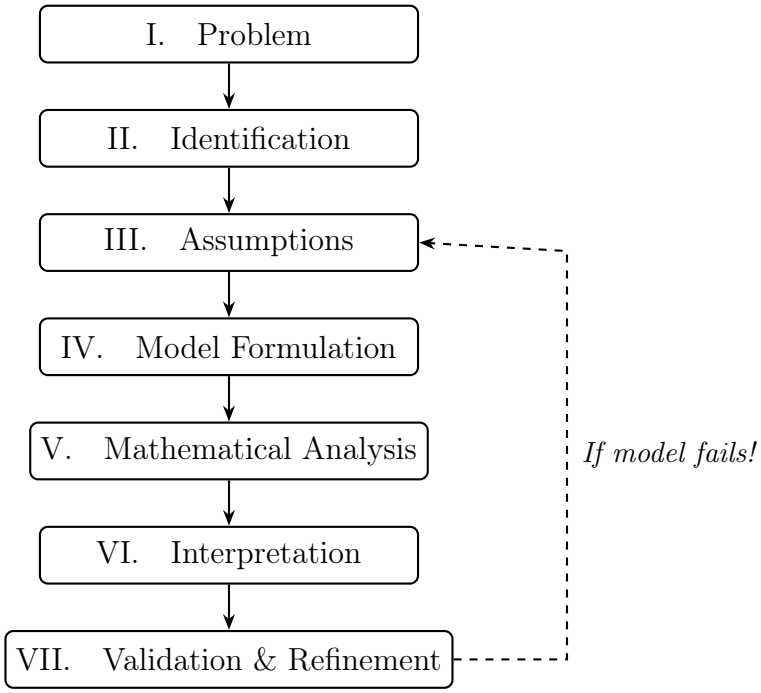
The roots of mathematical modelling begin from the ancient cultures where Babylonians, Egyptians, and Greeks used early mathematical techniques to solve practical problems such as land measurement, construction and architecture. Apart from those, scholars such as Al-Khwarizmi developed algebra, and innovations in trigonometry; Aryabhata introduced zero and decimal systems — all of which further reinforced the advancements in astronomy, optics, navigation and engineering. [14]

Further, the Renaissance and Enlightenment periods had remarkable advancements in mathematical modelling by the invention of calculus, enabling the modelling of dynamics and mechanical systems precisely in science and engineering. In the period of Industrial Revolution, mathematical modelling was further integrated into technological advancements, optimizing manufacturing processes and infrastructure design.

In the 20th century, the advent of computational power transformed mathematical modelling to the next level, allowing it to simulate complex systems and broaden its applications across different disciplines. As we move forward into the 21st century, the integration of mathematical modelling with data science, machine learning and artificial intelligence has the potential to address global challenges — ranging from climate change to healthcare advancements.

1.2 Flow Of Model

To build a mathematical model according to a problem we identified, does come up with certain steps [7]



So these are the steps, that we usually follow to model a problem. For a basic insight, let's consider population model as an example:

Population model or *Exponential growth model* are one of the major models in ecology. Let $N(t)$ be a population a time t. If each individual reproduces at rate r, then: [2]

$$\frac{dN}{dt} = r\,N,\ \underbrace{N(0) = N_0}_{\text{Intial Condition}} \tag{1}$$

Using the **variable separable method**:

$$\int \frac{dN}{N} = \int r\,dt \tag{2}$$

$$\ln(N) = rt + C \tag{3}$$

$$N = e^{rt+C} \tag{4}$$

$$N(t) = C\,e^{rt} \tag{5}$$

Applying initial conditions:

$$N(0) = N_0 \quad \Longrightarrow \quad C = N_0 \tag{6}$$

Hence:

$$\boxed{N(t) = N_0\,e^{rt}} \tag{7}$$

This is the way by which we model a system using mathematical tools, but throughout the article we mainly focus on **Heat Conduction Model**.

1.3 The Heat Conduction Model

The core element of every mathematical modelling is Differential Equations – equations that contain derivatives. Systems represented in these equations show change over certain parameters. [15]
Here we mainly discuss the **Transient Heat Conduction in a 1D Rod**. This is one of the major models, which is written in partial differential equations (PDEs), and generally referred to as the **Heat Equation**. The heat equation describes how energy is transferred from one body to another as the result of a temperature gradient. It describes a wide variety of phenomena over a broad range of scales, from heat diffusion in nanoscale materials to the macroscopic motion of atmospheric air [3]. Apart from physical applications, its references lead to applications in other disciplines such as finance (Black-Scholes Model). Recently, its applications cover Biology (Tissue-Damage Prediction), Fractional and Complex modelling (modelling anomalous diffusion in composite or heterogeneous materials with complex geometry), Geophysics (Study of the subsurface temperature distribution, thermal skin depth), and Nanofluids.

1.3.1 The Heat Equation

$$\frac{\partial T}{\partial t} = \alpha\,\frac{\partial^2 T}{\partial x^2} \tag{8}$$

[15]

where:

- $T(x,\,t)$ – temperature at position x and time t
- $\dfrac{\partial T}{\partial t}$ – rate of change of temperature with respect to time

- $\dfrac{\partial^2 T}{\partial x^2}$ – curvature of the temperature profile along the rod
- α – thermal diffusivity of the material $\left(\alpha = \dfrac{k}{\rho\, c_p}\right)$

Here, k is the thermal conductivity, ρ is the density, and c_p is the specific heat capacity of the material.

2 Basics and Classification

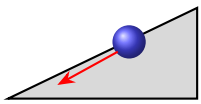
It is helpful to identify broad categories of models, so that classification of these models into these categories tells us immediately some of the essentials of their structure.One division between models is based on the type of outcome they predict.

2.1 Deterministic vs Stochastic Models

Deterministic models ignore random variation, and so always predict the same outcome from a given starting point. On the other hand, the model may be more statistical in nature and so may predict the distribution of possible outcomes. Such models are said to be stochastic. [8] [10]

Examples :

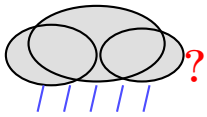
Deterministic



Ball rolling down a wedge

A ball rolling down a wedge follows a **fixed, predictable path**. Given the same initial conditions, the outcome is always the same.

Stochastic



Tomorrow's weather

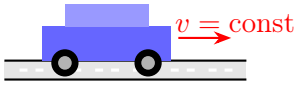
Tomorrow's weather is **uncertain and probabilistic**. Even with the same conditions, outcomes can differ each time.

2.2 Linear vs Non linear Autonomous models

According to the resulting equations, which may be algebraic, differential, or difference equations, being linear or non-linear, models are classified as linear or non-linear. A model is said to be linear when the variables are in the first power, which obeys superposition and can be solved analytically. In turn, non-linear models are those in which the variables have higher powers, which are more complex and can't be solved easily — they exhibit chaos! [8]

Examples :

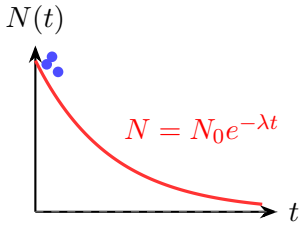
Linear



Car at constant speed

Governed by $x = vt$, a **first-power** relationship. Obeys superposition — *double the time, double the distance.* Can be solved analytically.

Non-Linear



Exponential decay of nuclear particle

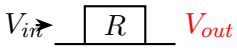
Governed by $\frac{dN}{dt} = -\lambda N$, a **non-linear** relationship in behaviour. The rate depends on the current quantity — *the more you have, the faster it decays.* Exhibits complex dynamics.

2.3 Statics vs Dynamic Models

In a static system, time doesn't play any part and hence the variables and the relationships describing the system are time independent In contrast dynamic systems time plays an important role. [8]

Examples :

Static

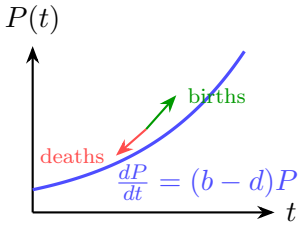


$V_{out} = I \cdot R$

Resistor Circuit

Output voltage V_{out} depends **strictly on the current input** — no memory of past states. The model captures a snapshot in time, with no evolution.

Dynamic



Population Model

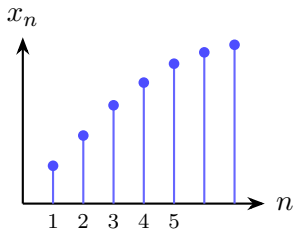
The future population **depends on current birth and death rates** — the model evolves over time. Past states directly influence future outcomes.

2.4 Continuous vs Discrete Models

A simple definition for a discrete and continuous models is that the dependant variable takes a certain discrete elements and gives discrete outputs which are characterized as discrete model, in contrast continuous models are based on continuous variables. [8]

Examples :

Discrete

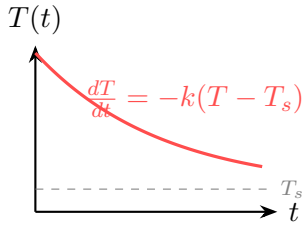


$x_{n+1} = r\,x_n$

Bank Account Balance

Defined at **distinct, separate points** in time — like monthly bank statements. The variable jumps from one step to the next with no values in between. Governed by *difference equations*.

Continuous



Newton's Law of Cooling

Defined **at every instant in time** — like temperature of a cooling object. The variable changes smoothly without any gaps. Governed by *differential equations*.

2.5 Autonomous vs Non-Autonomous

Autonomous systems operate independently based on their current state. Their governing equations do not explicitly depend on time, enabling self-regulation:

$$\frac{dx}{dt} = f(x)$$

Conversely, non-autonomous systems depend on external inputs or explicit time factors, requiring external control for behavioral changes:

$$\frac{dx}{dt} = f(x, t)$$

2.6 Category of Heat Conduction Model

In the earlier sections, we discussed the classification of mathematical models. Based on this understanding, let us now categorise the **heat conduction model**.

Continous....

Classification Comparison: Standard vs Temperature-Dependent α

Category	Standard ($\alpha = \text{const}$)	Project Model ($\alpha = \alpha(T)$)
Deterministic	✓ Unique solution given same B.C & I.C	✓ Still unique — conditions fully determine the solution
Linear / Non-Linear	✓ **Linear** — T in first power only	× **Non-Linear** — $\alpha(T)$ couples T to its own diffusion
Dynamic	✓ Evolves over time t	✓ Still evolves over time t
Continuous	✓ Defined at every x and t	✓ Still defined at every x and t
Autonomous	✓ No explicit t on right-hand side	✓ No explicit t — unless $\alpha = \alpha(T, t)$
Solvability	✓ Analytical solution exists	× No general analytical solution — requires numerical methods
Superposition	✓ Holds — sum of solutions is a solution	× Fails — material controls its own diffusion

The key shift is **Linear** $\rightarrow$ **Non-Linear**. When $\alpha = \alpha(T)$, the thermal diffusivity is no longer a fixed constant — it varies with the local temperature. Hotter regions diffuse heat differently than cooler ones, making the equation a **Quasilinear PDE**. This single change breaks superposition, eliminates analytical solutions, and demands numerical treatment — which forms the core challenge and motivation of this project.

3 Dimensional Analysis And Approximation

Dimensional analysis is one of the major tools used in mathematics, physics, and engineering. It is a method by which the relationships between physical quantities are analyzed with respect to fundamental quantities and their dimensions, such as mass (M), length (L), time (T), temperature (θ or K), and electric current (A). It helps us understand how physical quantities are related and simplifies complex problems into simpler ones. [12]

Apart from that, dimensional analysis can be used to determine whether an equation is physically meaningful, derive relationships between variables, and convert units from one system to another (for example, from the SI system $(\mathrm{kg}, \mathrm{m}, \mathrm{s})$ to the CGS system $(\mathrm{g}, \mathrm{cm}, \mathrm{s})$).

When we come to dimensional analysis, we must have knowledge of base quantities and derived quantities [16]. **Base quantities** are the fundamental physical quantities that are independent of one another and cannot be expressed in terms of other quantities. They serve as the foundation for measurement systems and dimensional analysis.

Derived quantities are the quantities obtained from base quantities through mathematical relationships such as multiplication, division, or powers. Their dimensions are expressed using combinations of fundamental dimensions. Examples include velocity, force, pressure, density, and energy. For example:

$$Velocity = \frac{Length}{Time} = [LT^{-2}]$$

One of the most important principles in dimensional analysis is **dimensional homogeneity [16]**. *Simply we could say that an apple can be added with another apple not an orange.* This principle ensures that an equation is physically consistent.
For example:

$$m = \frac{\vec{F}}{\vec{a}}$$

Let's check the dimensions

Dimension of $\vec{F} = [MLT^{-2}]$

Dimension of $\vec{a} = [LT^{-2}]$

Dimension of $m = \frac{[MLT^{-2}]}{[LT^{-2}]} = [M]$ *(Thus the equation is dimensionally homogenous.)*

3.1 Linearization

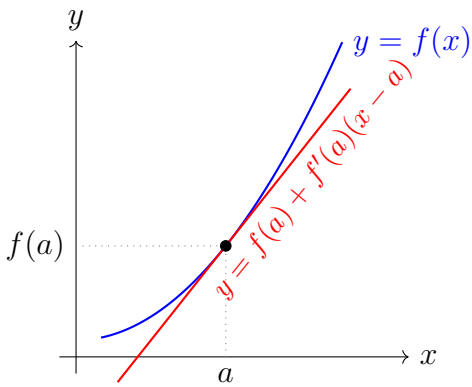
We often encounter difficulties while finding exact solutions for problems. If that is so, we often use **approximate** solutions for such cases, within some tolerance. Approximation is the process of finding a value which is close to the correct result. In mathematical modelling, approximation is necessary because: [13]

- Sometimes the data may be incomplete
- Presence of complex equations

- The exact values may not improve practical decisions

Here we are going to deal with linear approximation which is called **Linearization** by using the concept of tangent line.

Let f be a function differentiable at the point $x = a$. The equation of the tangent line to the graph of f at the point $x = a$ is $y = f(a) + f'(a)(x - a)$.



Definition [13]. The *linearization*, or *linear approximation*, of the function f near point $x = a$ is the linear function $L(x) = f(a) + f'(a)(x - a)$.

$f(x) \approx L(x)$ near $x = a$.

Example: Find the linear approximation to f(x) = sinx near x = 0

Solution. The linear approximation is the function $L(x) = f(a) + f'(a)(x - a)$, where $f(x) = \sin x$ and $a = 0$.

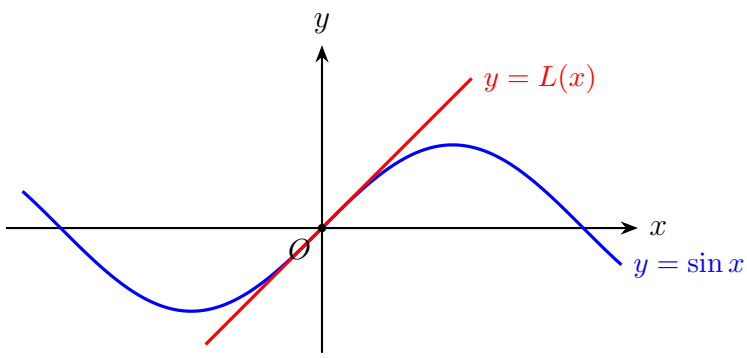
Since

$$f'(0) = f'(x)\Big|_{x=0} = (\sin x)'\Big|_{x=0} = \cos x\Big|_{x=0} = \cos 0 = 1$$

and $f(0) = \sin 0 = 0$, we find

$$L(x) = 0 + 1 \cdot (x - 0) \iff L(x) = x.$$

The linear function approximating $f(x) = \sin x$ near $x = 0$ is $L(x) = x$:



4 Mathematical Formulation of Transient Heat Conduction

As we discussed earlier, our main focus is on the heat conduction model. With reference to Section 1.2, before modeling a physical system, we need to identify the problem. Here, we are considering the heat flow through a 1D conductor; but before that, what is heat flow? [6] Heat energy is caused by the thermal agitation of molecular matter. There are basically two ways of transferring this energy: **Conduction** and **Convection**. *Conduction* results from collisions of neighboring molecules in which the kinetic energy of vibration of one molecule is transferred to its nearest neighbor. Thermal energy is thus spread by conduction even if the molecules themselves do not move their location appreciably. In addition, if a vibrating molecule moves from one region to another, it takes its thermal energy with it. This type of movement of thermal energy is called *Convection*. To begin simpler and as our main focus too, we consider conduction here.

4.1 Derivation of the Conduction of Heat in a One-Dimensional Rod

We begin by considering a rod of constant cross-sectional area A, oriented in the x-direction, where $x \in [0, L]$ [6, 1]. For the purpose of modeling we introduce amount of thermal energy per unit volume as a unknown variable and call it the **thermal energy density**:

$$\boxed{e(x,t) \equiv \text{thermal energy density.}}$$

We assume that all thermal quantities are constant across section; the rod is 1D. The simplest way is to insulate the lateral surface for avoiding the loss of heat through. The dependence on x and t corresponds to a situation in which the rod is not uniformly heated; the thermal energy density varies from one cross section to another.

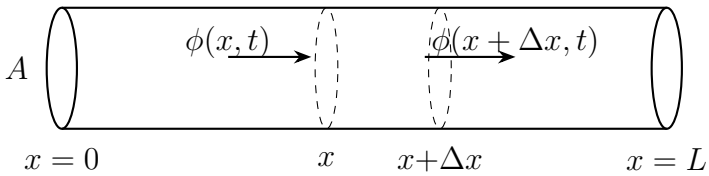


Figure 1: One-dimensional rod with heat energy flowing into and out of a thin slice.

Heat Energy, here we consider a thin slice of the rod $[x\ ,\ x+\Delta x]$ as illustrated in Figure 1 If the thermal energy density is constant throughout the volume then total energy (In general, energy density isn't constant but as Δx is extremelly small):

$$\boxed{\text{heat energy.} = e(x,t)A\Delta x}$$

Box 1 : Law of conservation of energy

According to the **Law of Conservation of energy** [1] the heat energy between $[x\ ,\ x+\Delta x]$ changes in time due only to heat energy flowing across the edges and heat energy that generated inside (due to positive or negative sources of heat energy). Then;

rate of change of heat energy in time	=	heat energy flowing across boundaries per unit time	+	heat energy generated inside per unit time.

Box 2 : Law of conservation of energy

This is called **conservation of heat energy**. For the small size, the rate of change of heat energy is

$$\frac{\partial}{\partial t}[e(x,t)A\Delta x], \tag{9}$$

where the partial derivative $\frac{\partial}{\partial t}$ is used because x is being held fixed.

Next we determine another parameter, **Heat flux (ϕ)**:

$$\boxed{\phi(x,t) = \text{the amount of thermal energy per unit time flowing to the right per unit surface area}}$$

The heat energy flowing per unit time across the boundaries of the slice is $\phi(x,t)A - \phi(x+\Delta x,t)A$. The sign of $\phi(x,t)$ determines the direction of heat flow:

- If $\phi(x,t) < 0$ — heat flows to the **left**
- If $\phi(x,t) > 0$ — heat flows to the **right**

As illustrated in Figure 1, when $\phi(x,t) > 0$ and $\phi(x+\Delta x,t) > 0$, the heat flow at x increases the energy in the slice, whereas the heat flow at $x+\Delta x$ decreases it.

heat sources. we also allow for internal sources of thermal energy:

$$\boxed{Q(x,t) = \text{heat energy per unit volume generated per unit time,}}$$

perhaps due to chemical reactions or electrical heating. $Q(x,t)$ is approximately constant in space for a thin slice, and thus the total thermal energy generated per unit time in the thin slice is approximately $Q(x,t)A\Delta x$.

Substituting the parameters in *Box 2*; rate of change of heat energy is due to thermal energy flowing across the boundaries and internal sources:

$$\frac{\partial}{\partial t}[e(x,t)A\Delta x] \approx \phi(x,t)A - \phi(x+\Delta x,t)A + Q(x,t)A\Delta x. \tag{10}$$

Equation 10, assumes various quantities to be approximately constant over the small slice, and becomes increasingly accurate as $\Delta x \to 0$. Directly taking the limit $\Delta x \to 0$ gives $0 = 0$, which is uninformative. Instead, we first divide by Δx, and then take the limit $\Delta x \to 0$, to obtain:

$$\frac{\partial e}{\partial t} = \lim_{\Delta x \to 0} \frac{\phi(x,t) - \phi(x+\Delta x,t)}{\Delta x} + Q(x,t) \tag{11}$$

where the constant cross-sectional area has been canceled. We claim this result is exact (with no small errors), and hence we can replace $\approx$ by $=$ in *Equation 10*. In this limting process, $\Delta x \to 0, t$ is being fixed. Consequently, from the definition of of PDE,

$$\frac{\partial e}{\partial t} = -\frac{\partial \phi}{\partial x} + Q \tag{12}$$

We usually describe materials by their **Temperature**,

$$\boxed{T(x,t) = \text{Temperature}}$$

not by their thermal energy density. And there is another important term we consider, the **Specific heat**

> c = specific heat (the heat energy that must be supplied to a unit mass of a substance to raise its temperature one unit).

Note: *In general, the specific heat c of a material depends on temperature T. In the 1D rod case, c depends on position x, i.e., $c = c(x)$. However, assuming the rod is made of a single uniform material, we treat c as a constant throughout.*

Thermal energy. The thermal energy in a thin slice is $e(x,t)A\,\Delta x$, which is also the energy required to raise the temperature from a reference temperature 0 to $T(x,t)$. Since c is independent of temperature, the heat energy per unit mass is $c(x)T(x,t)$. We thus introduce the **mass density** $\rho(x)$:

$$\boxed{\rho(x) = \text{mass density (mass per unit volume)},}$$

as it's a uniform rod we take ρ instead; From *Box 1* we know total thermal energy in the slice, we equate that with $c(x)T(x,t).\rho A\Delta x$, so that

$$e(x,t)A\Delta x = cT(x,t).\rho A\Delta x$$

canceling $A\Delta x$;

$$e(x,t) = c\rho T(x,t) \tag{13}$$

Relating Eqn 13 with Eqn 12, then we get;

$$c\rho\frac{\partial T}{\partial t} = -\frac{\partial \phi}{\partial x} + Q \tag{14}$$

According to **Fourier's Law of Heat Conduction.** [11] The heat flux ϕ is proportional to the temperature gradient, and heat flows in the direction of decreasing temperature [2]:

$$\phi = -k_0\frac{\partial T}{\partial x} \tag{15}$$

where:

- ϕ — heat flux; heat energy flowing per unit time per unit surface area
- k_0 — thermal conductivity of the material $(W/m \cdot K)$
- $\dfrac{\partial T}{\partial x}$ — temperature gradient
- The negative sign ensures heat flows from **hot** $\rightarrow$ **cold**

If Eqn 15 is substituted into Eqn 14, then

$$c\rho\frac{\partial T}{\partial t} = \frac{\partial}{\partial x}(K_0\frac{\partial T}{\partial x}) + Q$$

In addition, there are no sources, let $Q = 0$ and c, ρ, K_0 are constants, then;

$$c\rho\frac{\partial T}{\partial t} = K_0\frac{\partial^2 T}{\partial x^2}$$

And finally we get;

$$\boxed{\frac{\partial T}{\partial t} = \alpha\frac{\partial^2 T}{\partial x^2}} \tag{16}$$

The Heat Equation

where $\alpha = \frac{K_0}{c\rho}$ is the **thermal diffusivity** of the material — a constant that controls how fast heat spreads through the rod.

4.1.1 Upgrade: Temperature-Dependent Thermal Diffusivity

The standard heat equation derived above assumes that α is a **fixed constant** — meaning the material properties k_0, ρ, and c do not change with temperature. This is a reasonable approximation for small temperature variations, but **in reality**, the thermal conductivity of most materials varies with temperature.

Physical Motivation. Consider heating a metal rod to very high temperatures. The atoms vibrate more intensely, which changes how efficiently heat is conducted through the lattice. This means k is not constant — it depends on the local temperature $T(x, t)$:

$$\boxed{k = k(T) \quad \text{and therefore} \quad \alpha = \alpha(T) = \frac{k(T)}{c\rho}}$$

Substituting $k(T)$ back into Equation 15, Fourier's law becomes:

$$\phi = -k(T)\frac{\partial T}{\partial x}$$

and substituting this back, we get the **upgraded heat equation**:

$$c\rho\,\frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left[k(T)\,\frac{\partial T}{\partial x}\right] \tag{17}$$

Dividing through by $c\rho$, this can be written compactly as:

$$\boxed{\frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left[\alpha(T)\,\frac{\partial T}{\partial x}\right]} \tag{18}$$

The Heat Equation with Temperature-Dependent Thermal Diffusivity

4.1.2 Intial Conditions

To completely define the heat conduction problem, the PDE alone is not sufficient. We need additional information about [6, 15, 9]:

- The **initial state** of the rod at $t = 0$
- The **behaviour at the boundaries** $x = 0$ and $x = L$ for all time $t > 0$

These are called **Initial Conditions (IC)** and **Boundary Conditions (BC)** respectively. Together with the PDE, they form a **well-posed problem** — meaning a unique solution exists. [6]

Initial Condition. The initial condition specifies the temperature distribution throughout the rod at time $t = 0$:

$$T(x,\, 0) = f(x), \qquad 0 \leq x \leq L \tag{19}$$

In the simplest case, the rod starts at a **uniform temperature** T_0:

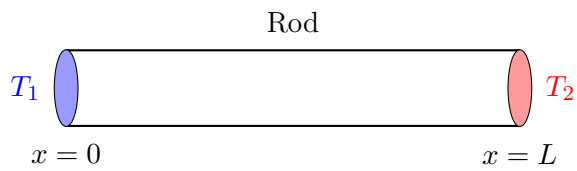
$$T(x,\, 0) = T_0$$

Boundary Conditions. The boundary conditions describe what happens at the two ends of the rod ($x = 0$ and $x = L$) for all $t > 0$. There are three classical types: [6]

1. Dirichlet Boundary Condition (First Kind)

The temperature is **prescribed** at the boundary. This is the most physically intuitive condition — we simply fix the temperature at the ends of the rod, like holding one end in ice and the other in a flame.

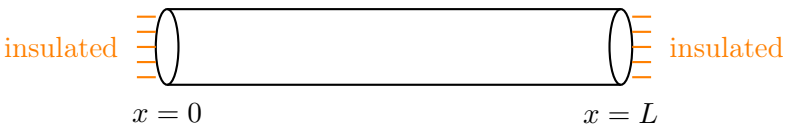
$$T(0,\, t) = T_1, \qquad T(L,\, t) = T_2, \qquad t > 0 \tag{20}$$



2. Neumann Boundary Condition (Second Kind)

The **heat flux** is prescribed at the boundary, meaning we control how much heat flows in or out. The special case of **zero flux** means the end is perfectly *insulated* — no heat escapes.

$$\left.\frac{\partial T}{\partial x}\right|_{x=0} = 0, \qquad \left.\frac{\partial T}{\partial x}\right|_{x=L} = 0, \qquad t > 0 \tag{21}$$



3. Robin Boundary Condition (Third Kind)

A **combination** of Dirichlet and Neumann — the heat flux at the boundary is proportional to the difference between the surface temperature and the surrounding (ambient) temperature T_∞. This is physically described by *Newton's Law of Cooling*:

$$-k \left.\frac{\partial T}{\partial x}\right|_{x=L} = h\,(T - T_\infty), \qquad t > 0 \tag{22}$$

where h is the **heat transfer coefficient** $(W/m^2 \cdot K)$ and T_∞ is the ambient temperature. The rod naturally loses more heat when it is hotter than its surroundings.

Summary of Boundary Conditions

Type	Condition	Physical Meaning
Dirichlet	$T = T_1$ at boundary	Fixed temperature
Neumann	$\frac{\partial T}{\partial x} = 0$ at boundary	Insulated / fixed flux
Robin	$-k\frac{\partial T}{\partial x} = h(T - T_\infty)$	Convective heat loss

In this project, we adopt **Dirichlet boundary conditions** with a uniform initial temperature, forming the complete problem statement:

$$\boxed{\begin{aligned} &\frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left[\alpha(T)\,\frac{\partial T}{\partial x}\right], \qquad 0 < x < L, \quad t > 0 \\ &T(x,\,0) = T_0 \qquad \text{(Initial Condition)} \\ &T(0,\,t) = T_1, \quad T(L,\,t) = T_2 \qquad \text{(Boundary Conditions)} \end{aligned}}$$

This is the complete mathematical model that will be solved numerically in the following sections.

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